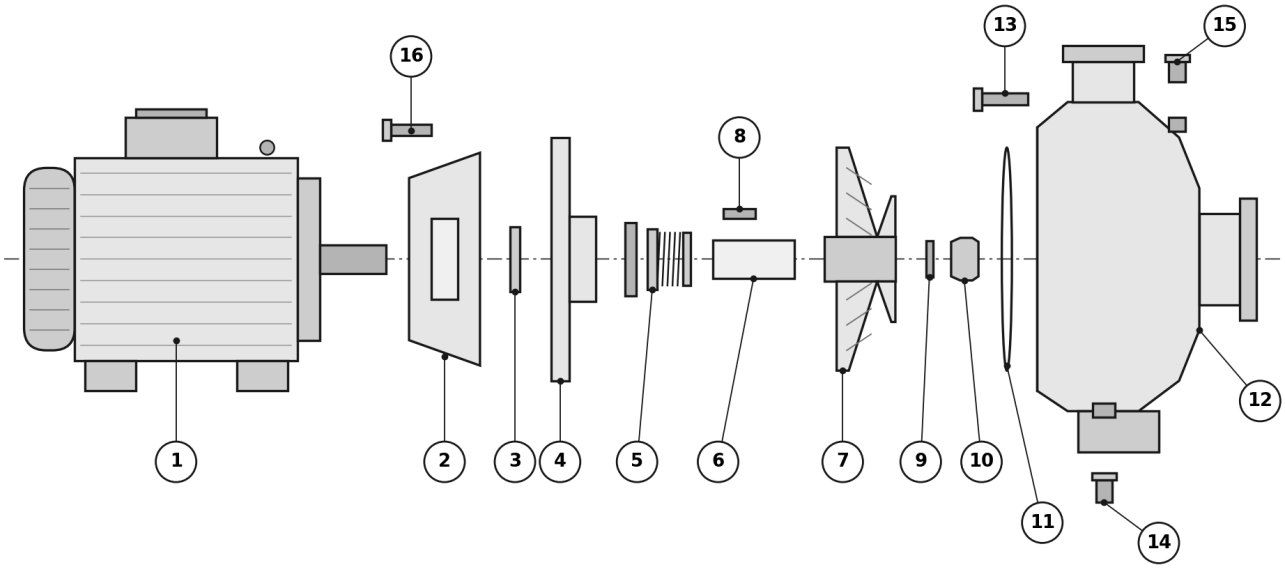
SHCHYGOLEVA INDUSTRIAL PUMPS

# Maintenance Manual

## SIP 80-200 Close-Coupled Centrifugal Water Pump

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**Read this manual before you do maintenance on the pump.**

Obey all safety instructions. Keep this manual near the pump.

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Portfolio sample: the product, company and part numbers are fictional.
Safety symbols are ISO 7010-style illustrations.

**Fictional equipment | Portfolio demonstration | Not for operational use**

Prepared by Scott Harmer

# Contents

**1 Introduction** 3

**2 Safety** 4

**3 Description** 5

**4 Maintenance schedule** 7

**5 Maintenance procedures** 8

5.1 Isolate the pump 8

5.2 Do the daily inspection 9

5.3 Lubricate the motor bearings 10

5.4 Replace the mechanical seal 11

5.5 Return the pump to service 14

## List of figures

Figure 1: Exploded view of the SIP 80-200 pump 6

Figure 2: Exploded view of the mechanical seal (item 5) 7

## List of tables

Table 1: Signal words 3

Table 2: Safety symbols 4

Table 3: Personal protective equipment 4

Table 4: Technical data 5

Table 5: Parts list 6

Table 6: Mechanical seal parts 7

Table 7: Maintenance schedule 8

Table 8: Personnel and equipment for procedure 5.1 8

Table 9: Personnel and equipment for procedure 5.2 9

Table 10: Personnel, equipment and consumables for procedure 5.3 10

Table 11: Personnel, equipment and consumables for procedure 5.4 11

Table 12: Torque values 11

Table 13: Personnel and equipment for procedure 5.5 14

# 1 Introduction

## 1.1 Purpose

This manual gives the maintenance data for the SIP 80-200 centrifugal water pump. It contains the safety instructions, the parts data, the maintenance schedule and the maintenance procedures.

## 1.2 Applicability

This manual applies to SIP 80-200 pumps with serial numbers 21000 and higher. The pump serial number is on the data plate on the pump casing.

## 1.3 Personnel

Only trained and approved personnel must do the procedures in this manual. Each procedure gives the necessary personnel:

- **Operator:** a person who is trained to operate the pump.
- **Maintenance technician:** a person who is trained to do mechanical maintenance on pumps.
- **Authorised electrician:** a qualified electrician who is approved to isolate and connect electrical equipment.

In procedures with more than one person, each step starts with the person who does the step: **Operator**, **Technician** (maintenance technician) or **Electrician** (authorised electrician).

## 1.4 Safety messages

This manual uses the safety messages in Table 1. Each message comes before the step to which it applies. Read each message fully before you do the step.

**Table 1: Signal words**

| Signal word | Meaning |
|---|---|
| DANGER | Shows a hazardous situation that will cause death or serious injury if you do not avoid it. |
| WARNING | Shows a hazardous situation that can cause death or serious injury if you do not avoid it. |
| CAUTION | Shows a hazardous situation that can cause minor or moderate injury if you do not avoid it. |
| NOTICE | Shows a risk of damage to the equipment or to property. A notice is not about personal injury. |
| NOTE | Gives information that helps you to do the procedure. A note does not contain a safety instruction. |

Each safety message tells you the hazard, the consequence and how to avoid the hazard.

This manual follows principles of Simplified Technical English (ASD-STE100) for clear instructions for all readers, including readers who do not speak English as their first language.

# 2 Safety

## 2.1 General safety rules

Obey these rules for all work on the pump:

- Read this manual before you do work on the pump.
- Obey your local health and safety regulations.
- Isolate the pump before you do maintenance (procedure 5.1). Use the lockout and tagout procedure of your site.
- Do not touch the pump until it is cool. The pump surfaces can be hotter than 60 °C.
- Drain the pump and release the pressure before you open it.
- Use lifting equipment to lift the motor assembly. It weighs approximately 120 kg.
- Wear the personal protective equipment in Table 3.
- Do not operate the pump with the guards or covers removed.
- Do not operate the pump if it does not contain water.

## 2.2 Safety symbols

This manual uses the safety symbols in Table 2.

**Table 2: Safety symbols**

| Symbol | Name | Meaning |
|---|---|---|
|  | **General warning** | Shows a hazard that can cause injury. |
|  | **Electricity** | The motor and the terminal box contain dangerous voltages. |
|  | **Hot surface** | The casing and the pipes can be hotter than 60 °C. |
|  | **Do not touch** | Do not touch the part while the pump operates. |
|  | **Disconnect before maintenance** | Isolate the electrical supply before you do maintenance. |

## 2.3 Personal protective equipment

Wear the personal protective equipment (PPE) in Table 3 when you do maintenance on the pump.

**Table 3: Personal protective equipment**

| Symbol | PPE | When to use |
|---|---|---|
|  | **Safety footwear** | For all maintenance work. |
|  | **Protective gloves** | When you touch parts, water or grease. |
|  | **Eye protection** | When you drain the pump, apply grease or use cleaning fluids. |
|  | **Read the manual** | Before you start each procedure. |

# 3 Description

## 3.1 General

The SIP 80-200 is a close-coupled, end-suction centrifugal pump. The impeller is installed directly on the motor shaft. Thus, the pump has no coupling and no separate bearing housing.

Water flows into the casing through the suction nozzle. The impeller turns and moves the water to the discharge nozzle. A mechanical seal prevents leakage where the shaft goes through the backplate.

The pump has a back pull-out design. You can remove the motor, the impeller and the mechanical seal as one assembly. The casing stays connected to the pipes.

## 3.2 Technical data

**Table 4: Technical data**

| Item | Value |
|---|---|
| Pump type | Close-coupled, end-suction centrifugal pump |
| Rated flow | 80 m³/h |
| Rated head | 32 m |
| Speed | 2950 rpm |
| Motor | 15 kW, 400 V, 50 Hz, 3-phase, efficiency class IE3 |
| Motor bearings | Drive end 6309 C3; non-drive end 6209 C3 (both can be lubricated) |
| Maximum water temperature | 80 °C |
| Maximum working pressure | 10 bar |
| Mechanical seal | 35 mm, carbon and silicon carbide faces, EPDM elastomers |
| Mass of the pump | Approximately 160 kg |
| Mass of the motor assembly | Approximately 120 kg |

## 3.3 Illustrated parts list

Figure 1 shows the pump parts. The item numbers in Figure 1 agree with the item numbers in Table 5 and in the procedures.

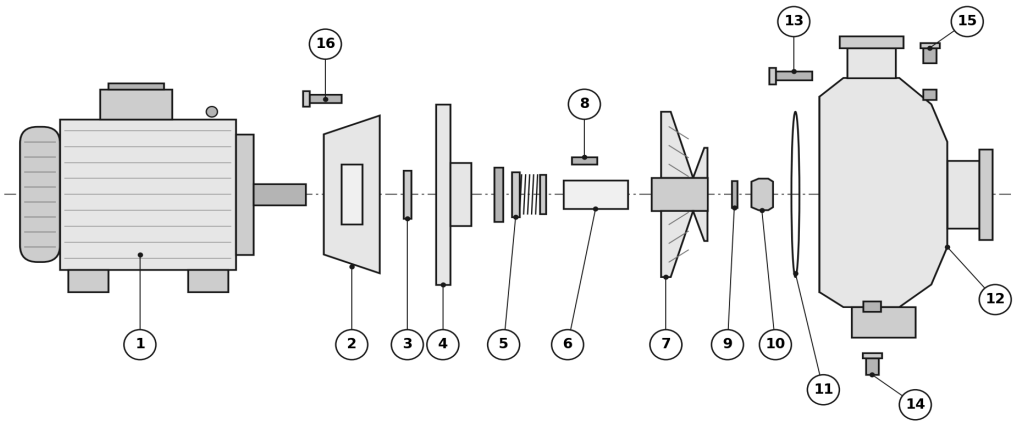



**Figure 1: Exploded view of the SIP 80-200 pump**

**Table 5: Parts list**

| Item | Part number | Description | Qty | Spare |
|---|---|---|---|---|
| 1 | SIP-80-101 | Electric motor, 15 kW, 2-pole, 400 V | 1 | |
| 2 | SIP-80-102 | Motor adapter | 1 | |
| 3 | SIP-80-103 | Water slinger | 1 | R |
| 4 | SIP-80-104 | Backplate (seal housing) | 1 | |
| 5 | SIP-80-105 | Mechanical seal, 35 mm, complete (see Figure 2) | 1 | R |
| 6 | SIP-80-106 | Shaft sleeve | 1 | W |
| 7 | SIP-80-107 | Impeller, 210 mm | 1 | W |
| 8 | SIP-80-108 | Impeller key | 1 | |
| 9 | SIP-80-109 | Impeller washer | 1 | |
| 10 | SIP-80-110 | Impeller nut, M16, self-locking | 1 | R |
| 11 | SIP-80-111 | Casing O-ring, EPDM | 1 | R |
| 12 | SIP-80-112 | Pump casing | 1 | |
| 13 | SIP-80-113 | Casing bolt, M12 × 40 | 8 | |
| 14 | SIP-80-114 | Drain plug, G 1/2, with seal | 1 | |
| 15 | SIP-80-115 | Vent plug, G 1/4, with seal | 1 | |
| 16 | SIP-80-116 | Adapter bolt, M10 × 30 | 4 | |

Spare: R = replace at each disassembly. W = wear part; examine at each disassembly and replace if necessary.

### How to order spare parts

To order spare parts, give this data to Shchygoleva Industrial Pumps:

- The pump model and the serial number. These are on the data plate on the pump casing.
- The item number and the part number from Table 5.
- The quantity of each part.

### 3.4 Mechanical seal parts

The mechanical seal (item 5) has a stationary part and a rotating part. You must replace the two parts together. Figure 2 shows the seal parts.

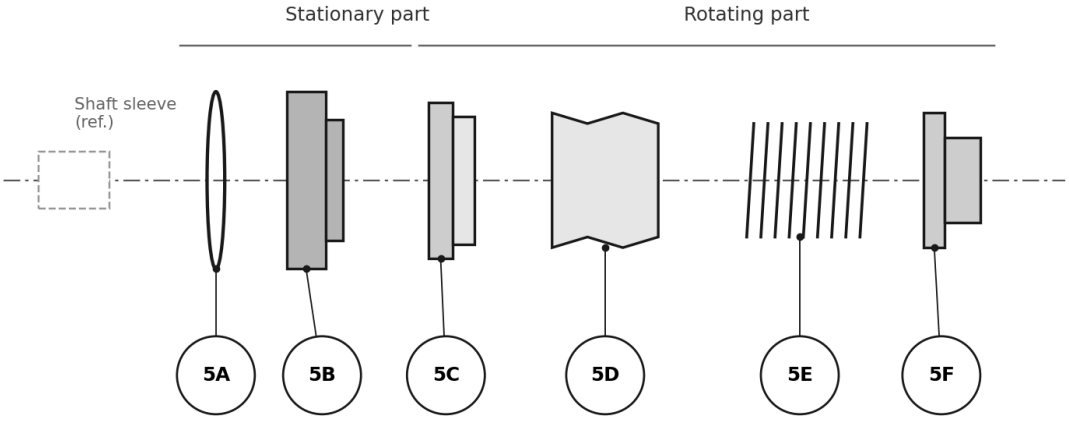



**Figure 2: Exploded view of the mechanical seal (item 5)**

**Table 6: Mechanical seal parts**

| Item | Description | Function |
|---|---|---|
| 5A | Seat O-ring | Seals the stationary seat in the backplate. |
| 5B | Stationary seat (silicon carbide) | Gives the stationary sealing face. |
| 5C | Rotating face (carbon) | Turns with the shaft against the stationary seat. |
| 5D | Bellows (EPDM) | Seals the rotating face on the shaft sleeve. |
| 5E | Spring | Pushes the rotating face against the stationary seat. |
| 5F | Spring retainer | Holds the spring in position on the shaft sleeve. |

## 4 Maintenance schedule

Do the maintenance tasks in Table 7 at the specified intervals. The intervals are for normal operation: clean water at 80 °C or less, and 16 hours or less of operation each day. If the pump operates in hard conditions, do the tasks more frequently. Hard conditions include dirty water, high temperatures and continuous operation.

Record each task in the maintenance log of the pump.

**Table 7: Maintenance schedule**

| Interval | Task | Procedure | Personnel |
|---|---|---|---|
| Daily | Do the daily inspection. | 5.2 | Operator |
| Weekly | Clean the motor fan cover and the motor cooling fins. | · | Operator |
| Monthly | Measure the vibration at the motor bearings. Record the values. | Site procedure | Maintenance technician |
| Monthly | Examine the pipe supports and the casing bolts for looseness. | · | Maintenance technician |
| Every 2000 operating hours | Lubricate the motor bearings. | 5.3 | Maintenance technician |
| Every 12 months | Measure the insulation resistance of the motor windings. | Site procedure | Authorised electrician |
| Every 12 months, or if the seal leaks | Replace the mechanical seal and the casing O-ring. | 5.4 | Maintenance technician |
| Every 12 months | Examine the impeller and the shaft sleeve for wear. | 5.4 | Maintenance technician |

# 5 Maintenance procedures

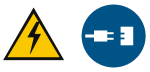
## 5.1 Isolate the pump

Do this procedure before all maintenance tasks that need the pump to be stopped.

**Table 8: Personnel and equipment for procedure 5.1**

| Personnel | Equipment |
|---|---|
| 1 maintenance technician<br>1 authorised electrician | Personal lock and danger tag<br>Valve locks or valve tags<br>Container, 20 litres<br>Spanner, 22 mm |

**! DANGER**



**Hazardous voltage in the motor and the terminal box.**
Contact with live electrical parts will cause death or serious injury.
» Make sure that an authorised electrician isolates the electrical power supply.
» Lock and tag the main isolator switch before you start work.

1. **Technician:** Tell the operators that you will stop the pump.
2. **Technician:** Push the STOP button on the control panel.
3. **Electrician:** Set the main isolator switch of the pump to OFF.
4. **Electrician:** Attach your personal lock and a danger tag to the main isolator switch.
5. **Technician:** Attach your personal lock to the main isolator switch.
6. **Technician:** Push the START button on the control panel. Make sure that the pump does not start.
7. **Technician:** Close the suction valve.
8. **Technician:** Close the discharge valve.
9. **Technician:** Attach a lock or a tag to each valve.

**WARNING**



**Hot water under pressure in the pump casing.**
The water can be at 80 °C and 10 bar. It can cause serious burns and eye injuries.
» Let the casing become cooler than 40 °C before you drain it.
» Release the pressure through the vent plug before you remove the drain plug.
» Wear eye protection and protective gloves.

**NOTICE**

**Water from the drain plug.**
Water can cause damage to the motor and to electrical equipment near the pump.
» Put a container below the drain plug before you open the plug.

10. **Technician:** Make sure that the casing temperature is less than 40 °C.
11. **Technician:** Put the container below the drain plug (14).
12. **Technician:** Loosen the vent plug (15) by one turn. Let the pressure decrease.
13. **Technician:** Make sure that the pressure gauge on the discharge pipe shows 0 bar.
14. **Technician:** Remove the drain plug (14). Let the water drain into the container.
15. **Technician:** Remove the vent plug (15).

## 5.2 Do the daily inspection

Do this inspection while the pump operates. Do not remove the covers or guards.

**Table 9: Personnel and equipment for procedure 5.2**

| Personnel | Equipment |
|---|---|
| 1 operator | Daily log sheet |

**WARNING**



**Rotating parts and hot surfaces.**
Contact with the shaft, the motor fan or the casing can cause serious injury or burns.
» Do not touch the pump while it operates.
» Do not remove the covers or the guards.

1. Examine the area below the backplate (4) for water.
   a. If you see water drops from the backplate, replace the mechanical seal (procedure 5.4).
2. Listen to the pump and the motor.
   a. If you hear an unusual noise, stop the pump. Tell your supervisor.
3. Read the pressure gauge on the discharge pipe.
   a. Make sure that the pressure is between 3.0 bar and 3.4 bar.
   b. If the pressure is not in this range, tell your supervisor.
4. Examine the casing bolts (13), the drain plug (14) and the pipe flanges for leaks.
5. Make sure that the motor fan cover is clean and that air can flow through it.
6. Record the results in the daily log.

## 5.3 Lubricate the motor bearings

Do this procedure every 2000 operating hours. The motor has a grease nipple at each bearing: one at the drive end and one at the non-drive end.

**Table 10: Personnel, equipment and consumables for procedure 5.3**

| Personnel | Equipment | Consumables |
|---|---|---|
| 1 maintenance technician<br>1 authorised electrician | Grease gun with a calibrated output<br>Lint-free cloth | Lithium-complex grease, NLGI grade 2: 15 g for the drive end, 10 g for the non-drive end |

**NOTICE**

**Incorrect grease.**
Incorrect grease or a mixture of grease types can cause damage to the motor bearings.
» Use only lithium-complex grease, NLGI grade 2. Do not mix different types of grease.

**NOTICE**

**Too much grease.**
Too much grease causes high bearing temperatures and can cause damage to the bearings.
» Apply only the specified quantity of grease to each bearing.

1. **Technician:** Isolate the electrical supply. Do steps 1 to 6 of procedure 5.1.
2. **Technician:** Clean the two grease nipples on the motor (1) with a lint-free cloth.
3. **Technician:** Remove the grease outlet plug at each bearing.
4. **Technician:** Apply 15 g of grease to the drive-end grease nipple.
5. **Technician:** Apply 10 g of grease to the non-drive-end grease nipple.
6. **Technician:** Clean the grease nipples again.
7. **Technician:** Start the pump. Do steps 8 to 12 of procedure 5.5.
8. **Technician:** Let the pump operate for 30 minutes. Unwanted grease comes out of the outlets.
9. **Technician:** Install the grease outlet plugs.
10. **Technician:** Record the task in the maintenance log.

## 5.4 Replace the mechanical seal

Do this procedure every 12 months, or if the mechanical seal leaks. The procedure also includes the inspection of the impeller and the shaft sleeve.

**Table 11: Personnel, equipment and consumables for procedure 5.4**

| Personnel | Equipment | Consumables and spares |
|---|---|---|
| 2 maintenance technicians<br>1 authorised electrician | Lifting equipment, 250 kg minimum<br>Socket spanners: 16 mm, 18 mm, 24 mm<br>Torque wrench, 20 to 100 N·m<br>Strap wrench<br>Impeller puller<br>Soft-face hammer | Mechanical seal (5)<br>Casing O-ring (11)<br>Impeller nut (10)<br>Soapy water<br>Isopropyl alcohol<br>Lint-free cloth |

**Table 12: Torque values**

| Item | Description | Torque |
|---|---|---|
| 10 | Impeller nut, M16 | 90 N·m |
| 13 | Casing bolts, M12 | 75 N·m |
| 14 | Drain plug, G 1/2 | 30 N·m |
| 16 | Adapter bolts, M10 | 45 N·m |

### Remove the motor assembly

**WARNING**



**Heavy load. The motor assembly weighs approximately 120 kg.**
A falling assembly can cause serious injury.
» Use lifting equipment with a capacity of 250 kg or more.
» Do not stand below the load.
» Wear safety footwear.

1. **Technician:** Isolate and drain the pump. Do procedure 5.1.
2. **Electrician:** Disconnect the motor cables from the terminal box.
3. **Technician:** Attach the lifting equipment to the lifting eye of the motor (1).
4. **Technician:** Remove the eight casing bolts (13).
5. **Technician:** Pull the motor assembly straight back from the pump casing (12).
6. **Technician:** Put the motor assembly on a clean work bench. Support the motor so that it cannot move.
7. **Technician:** Remove the casing O-ring (11). Discard the O-ring.

### Remove the impeller and the mechanical seal

**NOTICE**

**Impact on the impeller or the shaft.**
A steel hammer can cause damage to the motor bearings and the shaft.
» Use only a soft-face hammer. Do not hit the impeller or the shaft with a steel hammer.

1. **Technician:** Hold the impeller (7) with the strap wrench.
2. **Technician:** Remove the impeller nut (10) and the impeller washer (9). Discard the impeller nut.
3. **Technician:** Remove the impeller (7) with the impeller puller.
4. **Technician:** Remove the impeller key (8).
5. **Technician:** Remove the rotating part of the mechanical seal (5C to 5F) from the shaft sleeve (6).
6. **Technician:** Remove the shaft sleeve (6).
7. **Technician:** Remove the four adapter bolts (16).
8. **Technician:** Remove the backplate (4) from the motor adapter (2).
9. **Technician:** Push the stationary seat (5B) and the seat O-ring (5A) out of the backplate. Discard the seal parts.

### Examine the parts

1. **Technician:** Examine the shaft sleeve (6) for grooves and scratches.
   a. If the surface has grooves, replace the shaft sleeve.
2. **Technician:** Examine the impeller (7) for cracks, erosion and corrosion.
   a. If the impeller has cracks or erosion outside approved limits, replace the impeller.
3. **Technician:** Clean the seal bore in the backplate (4) with a lint-free cloth.
4. **Technician:** Examine the seal bore for scratches.

### Install the new mechanical seal and the impeller

**NOTICE**

**Contamination of the new seal.**
Dirt on the sealing faces causes leaks. Oil and grease cause damage to the EPDM parts.
» Do not touch the sealing faces with your fingers.
» Do not use oil or grease on the EPDM parts.

**NOTE**

Use soapy water as a lubricant for the O-ring and the bellows. Soapy water lets the parts move easily during installation.

1. **Technician:** Apply soapy water to the new seat O-ring (5A).
2. **Technician:** Install the seat O-ring (5A) on the new stationary seat (5B).
3. **Technician:** Push the stationary seat (5B) fully into the seal bore of the backplate (4). Push equally with your thumbs.
4. **Technician:** Clean the sealing face of the stationary seat (5B) with isopropyl alcohol and a lint-free cloth.
5. **Technician:** Install the backplate (4) on the motor adapter (2).
6. **Technician:** Install the four adapter bolts (16). Tighten the bolts to 45 N·m.
7. **Technician:** Install the shaft sleeve (6) on the motor shaft.
8. **Technician:** Clean the sealing face of the rotating face (5C) with isopropyl alcohol and a lint-free cloth.
9. **Technician:** Apply soapy water to the inner surface of the bellows (5D).
10. **Technician:** Push the rotating part (5C to 5F) onto the shaft sleeve until the two sealing faces touch.
11. **Technician:** Install the impeller key (8).
12. **Technician:** Install the impeller (7) on the shaft.
13. **Technician:** Install the impeller washer (9) and a new impeller nut (10).
14. **Technician:** Hold the impeller with the strap wrench. Tighten the impeller nut to 90 N·m.
15. **Technician:** Turn the impeller by hand. Make sure that it turns freely.

### Install the motor assembly

**CAUTION**



**Pinch points between the motor assembly and the pump casing.**
Your hands can be caught between the parts. This can cause minor or moderate injury.
» Keep your hands away from the flanges when you push the assembly into the casing.
» Wear protective gloves.

1. **Technician:** Install a new casing O-ring (11) in the groove of the pump casing (12).
2. **Technician:** Lift the motor assembly with the lifting equipment.
3. **Technician:** Push the motor assembly straight into the pump casing (12).
4. **Technician:** Install the eight casing bolts (13).
5. **Technician:** Tighten the casing bolts in a diagonal sequence to 75 N·m.
6. **Technician:** Remove the lifting equipment.
7. **Electrician:** Connect the motor cables to the terminal box.
8. **Technician:** Return the pump to service. Do procedure 5.5.

## 5.5 Return the pump to service

Do this procedure after each maintenance task that needed procedure 5.1.

**Table 13: Personnel and equipment for procedure 5.5**

| Personnel | Equipment |
|---|---|
| 1 maintenance technician<br>1 authorised electrician | Torque wrench, 20 to 100 N·m<br>Socket spanner, 22 mm<br>Maintenance log |

**NOTICE**

**Dry operation.**
If the pump operates without water, the mechanical seal can be damaged in less than 1 minute.
» Fill the pump with water before you start it.

1. **Technician:** Install the drain plug (14). Tighten the drain plug to 30 N·m.
2. **Technician:** Install the vent plug (15). Do not tighten it.
3. **Technician:** Remove the locks or tags from the valves.
4. **Technician:** Open the suction valve slowly.
5. **Technician:** Let water flow out of the vent plug (15) until no more air comes out.
6. **Technician:** Tighten the vent plug (15).
7. **Technician:** Open the discharge valve to approximately 25%.

**WARNING**



**Rotating parts when the pump starts.**
Contact with rotating parts can cause serious injury.
» Make sure that all guards and covers are installed.
» Make sure that all persons are away from the pump before you start it.

8. **Technician:** Remove your personal lock from the main isolator switch.
9. **Electrician:** Remove your personal lock and the danger tag from the main isolator switch.
10. **Electrician:** Set the main isolator switch to ON.
11. **Technician:** Tell the operators that you will start the pump.
12. **Technician:** Push the START button on the control panel.
13. **Technician:** Open the discharge valve slowly to the fully open position.
14. **Technician:** Examine the area below the backplate (4) for leaks.

**NOTE**

A new mechanical seal can let a small quantity of water through for the first 10 minutes. If the leakage continues after 30 minutes, stop the pump and tell your supervisor.

15. **Technician:** Make sure that the discharge pressure is between 3.0 bar and 3.4 bar.
16. **Technician:** Record the work in the maintenance log of the pump.